DA User Guide

DA (short for *Digital Archive*) puts your photos and scanned documents in order. You copy them into one folder, and DA:

- sets exact duplicates aside, so you keep one copy of each photo,
- works out when each photo was taken, and renames it by that date (`photo_19980704_101500_ab12.jpg` was taken on 4 July 1998 at 10:15:00),
- files it in a folder for its year and month (`photos\organized\1998\1998-07\`).

DA never deletes anything. It asks before it moves anything, and it keeps a record of every change, so every step can be checked and most can be undone.

DA runs on Windows 10 and 11.

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Words you'll see

Word	What it means
Archive	A folder DA looks after, for example `D:\jane-archive-199x`. You can have as many as you like, for example one for each person and each decade.
Intake	The folder inside an archive where you put new photos and documents: `intake\`. DA takes them from here.
Organized	Where DA files them: `photos\organized\` and `documents\organized\`, one folder per year and month.
Duplicate	An exact copy of a photo, byte for byte. DA keeps one and moves the others to `_duplicates-holding\`.
Sidecar	A small note DA keeps next to each file, named like the file plus `.meta.json`. It holds the date DA found, a fingerprint of the file, and any tags. Leave sidecars where they are.
Terminal	A window where you type commands. Windows calls it *PowerShell* or *Terminal*. DA's menu runs in one.
Command	Something you type in the terminal, like `da status`, then press **Enter**.

Getting started

1. Get DA

- **Download** DA.exe (the latest version), or get the file from whoever set DA up for you (for example on a USB stick).
- Your browser may ask whether to keep the file. Choose **Keep**.

2. Install it (once per computer)

1. Open **File Explorer** and go to the folder with `DA.exe` (usually **Downloads**).
2. Click in the **address bar** at the top of the window, type `powershell`, and press **Enter**. A terminal window opens in that folder.
3. Type this and press **Enter**: `powershell .\DA.exe install`
4. If Windows shows "*Windows protected your PC*", click **More info**, then **Run anyway**.

5. **Close the terminal window.** From now on, `da` works in any new terminal window.

DA is now at `C:\Scripts\DA\DA.exe`. To update DA later, download the new `DA.exe` and do these steps again; your archives aren't touched.

3. Create your first archive (once per archive)

1. Open a new terminal: press the **Windows key**, type `terminal`, press **Enter**.
2. Type this (with your own drive, name and years) and press **Enter**: `powershell da setup D:\jane-archive-199x --years 1990-1999`

What happens depends on the folder:

- **A new folder:** DA creates it, with everything inside it that an archive needs.
- **A folder you've already been copying photos into:** DA lists what's there and asks "*Make it an archive and move these N items into intake\?*". Type **y**. Everything moves into `intake\` with its names unchanged, and the move is written down so it can be undone.
- **A folder that's already an archive:** DA asks "*Use the existing archive?*", keeps its settings, and adds anything missing.

`--years 1990-1999` is optional. With it, a photo taken in another decade (say 2005) is left in `intake\` and listed for you, so you can move it to the right archive (see da split).

Your archive now looks like this:

```
D:\jane-archive-199x\
  Open DA.cmd           <- double-click this to open DA in this archive
  da-archive.yaml       <- this archive's settings
  intake\
    photos\             <- put photos here (any subfolders, any names)
    docs\               <- documents: PDF, Word, Excel
    scans\              <- what your scanner saves: PDFs and scanned pictures
  photos\organized\     <- DA files photos here, by year and month
  documents\organized\  <- and documents here
  output\               <- DA's records and reports
```

4. Open it

Double-click `Open DA.cmd` in the archive folder. DA's menu opens. That's all you need from here on.

The menu

```
DA - jane-archive-199x   D:\jane-archive-199x   years: 1990-1999
intake: 1,204 files waiting    organized: 5,873 photos, 12 documents    last run: 2026-10-13 2

 1 Process intake    2 Status    3 What needs review    4 Undo last set-aside
 5 Refresh index     6 Switch archive    7 New archive    U User guide    H Help    Q Quit
```

Type the number or letter, then press **Enter**.

Key	What it does
1 Process intake	Files what's in `intake\`. See da run.
2 Status	This archive at a glance: what's waiting, what's filed, the last run.
3 What needs review	What the last run left for you to look at, with the reasons.
4 Undo last set-aside	Puts the duplicates from the last run back where they were.
5 Refresh index	Updates DA's list of files, for example after you moved files yourself.
6 Switch archive	Open another archive on this computer.
7 New archive	Create a new archive, or turn a folder into one.
U User guide	Opens this guide in your web browser.
H Help	A short reminder of how DA works.
Q Quit	Leaves the menu. The window stays open for typing commands.

Typing `da` on its own in any terminal opens the menu too. Outside an archive, it lists the archives on this computer for you to pick from.

Common workflows

Workflow	What it's for	Menu
Add photos from a phone or camera	Get photos into intake	(File Explorer)
Add scanned documents	Get paper into the archive	(your scanner)
da run	File what's in intake	**1**
da report	See what's there and what needs you	**2**, **3**
Fix what needs review	Help DA with photos it couldn't date	**3**
Duplicates: put back or delete	Undo a set-aside, or free the space	**4**
da split	Move photos from another decade to the right archive	steps below
da find	Search by month, name or tag	steps below
da tag	Label files (`beach` , `grandma` , `taxes`)	steps below
Switch between archives	Work in another archive	**6**
Update DA	Get the newest version	steps below

`da run` is a command you can type. The other "da ..." names stand for a few commands or menu items used together.

Add photos from a phone, camera or memory card

1. **Open the archive's intake folder** in File Explorer: `D:\jane-archive-199x\intake\photos\` .
2. **Make a folder for this batch,** named so you'll recognise it, e.g. `phone-2024-summer` .
3. **Copy the photos into it:**
    - **iPhone:** connect it with its cable and unlock it. Tap **Trust** if it asks. In File Explorer, open **This PC > Apple iPhone > Internal Storage > DCIM** and copy the folders you want. (See iPhone HEIC photos about the photo format.)
    - **Android phone:** connect it, choose **File transfer** on the phone, then open **This PC > (your phone) > Internal storage > DCIM > Camera**.

    - **Camera or memory card:** put the card in the computer (or connect the camera), then open its drive and the **DCIM** folder.
4. **Copy, don't move,** until DA has filed them and you've checked the result.
5. Then process them.

Any folder names and file names are fine. DA works out the dates itself.

Add scanned documents

1. **Set your scanner to save into** `D:\jane-archive-199x\intake\scans\` (or copy the files there afterwards).
2. **PDF is best** for documents. Pictures (`.jpg` , `.png` , `.tif`) saved in `scans\` are treated as documents too, not as photos.
3. Word and Excel files and other PDFs go in `intake\docs\` .
4. Then process them. Documents are filed in `documents\organized\` by the date inside them (when they were scanned or created).

da run: file new photos and documents

Menu: 1 Process intake. DA lists the folders in `intake\` and how many files each holds:

```
intake\photos\
 #  folder                          files
 0  process all of intake\photos\  5,081
 1  phone-2024-summer\             1,204
 2  scans-from-grandma\              877
  number: open   0: process   text: filter   .. up   Enter: back
```

- Type a **number** to open that folder. A folder with no folders inside it is processed straight away.
- Type **0** to process the folder you're in, everything in it.
- Type some **text** (e.g. `grandma`) to show only the folders whose names contain it.
- Type **..** to go up a level, or just press **Enter** to go back.

DA then works through seven steps. Before each step that changes anything, it asks; type **y** to go ahead, or **n** to stop there (nothing after that point changes).

Step	What happens	Changes files?
1 Index	Lists everything in the archive	No
2 Hash	Fingerprints the photos that could be copies	No
3 Duplicates	Moves exact copies to `_duplicates-holding\`	**Yes, if you say y**
4 Preview	Works out each photo's date, new name and folder	Only writes the sidecar notes
5 Check	Shows how many will be filed, by year, and warns about anything odd	No
6 File	Renames and moves the photos into `organized`, then checks each one arrived intact	**Yes, if you say y**
7 Refresh	Updates the index	No

How long it takes: a few minutes for a few hundred photos; roughly an hour for every 50,000. It's fine to leave it running.

What stays in intake, on purpose: - **Photos with no date anywhere** (none inside the file, none in its name). See Fix what needs review. - **Photos from another decade** (with `years` set). See da split. - **Videos** (`.mov` , `.mp4`): DA doesn't file videos yet. - **Camera raw files** (`.CR2` , `.NEF` , ...) without a matching JPG. With a matching JPG, the raw file is filed right next to it.

Running it again is always safe. Files already filed aren't touched, and new ones are processed the same way.

From a terminal (handy for leaving a big run going overnight):

```
da run                                  # all of intake
da run intake\photos\phone-2024-summer  # one folder
da run --yes                            # don't ask; go straight through
da run E:\old-backup                    # a folder outside the archive: DA offers to copy it ir
```

da report: what's in the archive and what needs you

Menu: 2 Status and **3 What needs review.**

1. **2 Status:** the archive's name and years, what's waiting in intake, what's filed, the last run.

2. **3 What needs review:** what the last run left behind, grouped by reason, with examples:

    - *No date (in the file or its name)*
    - *Taken outside this archive's years*
    - *File name and camera date disagree* (DA used the date in the name)
    - *Duplicate inside the folder* (left for you to decide)

    The full list is saved as `review.csv` in that run's folder (DA shows where). It opens in Excel. 3. **More detail, from a terminal:** `powershell da db stats # how many photos and documents, by year da intake status # how far each intake folder has got da validate photos\organized # check what's already filed: names, folders, notes (nothing moves)` 4. **Every run's record** is kept in `output\runs\<date-time>-run\` : a summary, the duplicate list, and the preview and filing reports, file by file.

If you added or moved files yourself, choose **5 Refresh index** first so the numbers are current.

Fix what needs review

Choose **3 What needs review**, then for each reason:

- **No date:** DA couldn't find when the photo was taken. If you know (even roughly), put the date at the start of its name and run again:

    - `1998-07-04 Beach.jpg` (day known)
    - `1998-07 Beach.jpg` (month known)

    A year on its own (`1998 Beach.jpg`) isn't enough to file it, so it stays. - **Outside this archive's years:** it belongs in another archive. See da split. - **Name and camera date disagree:** usually the camera's clock was wrong. DA used the date in the name. If the name is the wrong one, rename the file in `organized` yourself. - **Duplicate inside the folder:** two identical files. DA leaves both; delete the one you don't want.

Duplicates: put back or delete

Exact copies are moved to `_duplicates-holding\<date>\` in the archive, in the same folders they came from.

- **Put them back: 4 Undo last set-aside**. Or, for an earlier run, from a terminal: `da intake undo "<run folder>\set-aside-moved.csv"` .
- **Delete them** to free the space, once you're happy with the result: delete the `_duplicates-holding` folder. Every file in it is an exact copy of one that was kept.

da split: move files to the right decade's archive

With `years` set (e.g. `1990-1999`), a photo taken in another decade stays in intake, listed as "*Taken outside this archive's years*". To move those to their own archive:

1. **List them: 3 What needs review**. Note where it saved `review.csv`.
2. **Create (or pick) the other archive:** `da setup D:\jane-archive-200x --years 2000-2009`.
3. **Move them there, with their sidecars.** In a terminal, with your own paths (this moves the 2000s ones; the `detail` column starts with the date, e.g. *Taken 2005-07-04, ...*): `powershell $review = "D:\jane-archive-199x\output\runs\<date-time>-run\review.csv" $to = "D:\jane-archive-200x\intake\photos\from-199x" New-Item -ItemType Directory -Force $to | Out-Null Import-Csv $review | Where-Object { $_.detail -like "Taken 200*" } | ForEach-Object { Move-Item -LiteralPath $_.file -Destination $to if (Test-Path -LiteralPath "$($_.file).meta.json") { Move-Item -LiteralPath "$($_.file).meta.json" -Destination $to } }` If two files have the same name, the second gives an error and stays where it was; move it by hand. A raw file (`.CR2`) beside a moved JPG isn't on the list; move it too.
4. **File them there:** open the other archive (its `Open DA.cmd`, or **6 Switch archive**) and choose **1 Process intake**, folder `from-199x`.

da find: search the archive

DA keeps an index of every file: its name, folder, date and tags. Choose **5 Refresh index** first if anything changed, then in a terminal in the archive:

- **Everything from one month** (by the folder it's filed in): `powershell da db query "SELECT file_name, folder FROM files WHERE area='photos' AND folder LIKE '2010\2010-07%'"`
- **By name:** `... WHERE file_name LIKE '%0704%'`. For the name a file had before DA renamed it: `... WHERE source_path LIKE '%Beach%'`.
- **Documents:** use `area='documents'` instead of `area='photos'`.
- **By tag** (see da tag): `da findit tags photos\organized -r beach -v`. Add `-r grandma` for both tags, `-x beach` to leave a tag out, or `--date-prefix 2010-07` for one month.

Or just browse: every photo is in `photos\organized\<year>\<year>-<month>\`, in date order.

da tag: label photos and documents

There's no tag command yet, so for now a tag is added by editing the file's sidecar:

1. **Open the sidecar** next to the photo, e.g. `photo_19980704_101500_ab12.jpg.meta.json`, with **Notepad** (right-click > **Open with** > **Notepad**).
2. **Find the** `"photoit"` **part** (for a document, `"scanit"`) and its `"tags": []` line. Put your tags inside the brackets, in quotes, with commas between them: `json "tags": ["beach", "grandma"],` Change

nothing else, and save.
3. **Choose 5 Refresh index**, so da find can see the new tags.

Switch between archives

- **6 Switch archive** in the menu lists every archive on this computer. Type its number.
- Or double-click that archive's `Open DA.cmd`.
- Or in a terminal: `da archives` lists them, and `da archives --open 2` opens number 2.

Update DA

Download the newest DA.exe and install it the same way as the first time (Getting started). Close any open DA windows first. Your archives aren't touched. To see which version you have: `da version`.

Less common workflows

Workflow	What it's for
iPhone Live Photos (.MOV files)	The short videos that come with iPhone photos
iPhone HEIC photos	The iPhone's photo format
Edit an archive's settings	Change its name, years or intake folders
Share an album with Google Photos	Let family see some photos online
Process a folder from somewhere else	A USB stick, an old backup, another drive
Start a new archive for another person or decade	One archive per person per decade
Use DA on another computer	Take an archive on a USB drive to another PC
Back up an archive	Keep a second copy safe
Check what's already filed	A check-up of `organized`
Do the run's steps one at a time	For a careful look between each step
Coming soon	What DA will do next

iPhone Live Photos (.MOV files)

An iPhone *Live Photo* is two files with the same name: the photo (`IMG_1234.HEIC` or `.JPG`) and a 2-3 second video (`IMG_1234.MOV`).

- **What DA does today:** it files the photo and leaves the `.MOV` in intake, because DA doesn't file videos yet. Nothing is lost; the `.MOV` just stays where it was.
- **To see how many are waiting: 2 Status**, or in a terminal: `da db query "SELECT COUNT(*) AS movies FROM files WHERE area='intake' AND extension='.mov'"` .
- **Don't want them?** On the iPhone, turn off **Live** (the round icon at the top of the camera) before taking photos.
- **Coming soon:** DA will pair each `.MOV` with its photo, so it's filed next to it with the same name, and find the ones already left behind. See Coming soon.

iPhone HEIC photos

iPhones save photos as **HEIC** (`IMG_1234.HEIC`): the same quality as JPG in about half the space.

- **DA files HEIC photos like any other,** using the date the iPhone stored in them.
- **To see them on Windows,** install the free **HEIF Image Extensions** from the Microsoft Store (open the **Microsoft Store** app and search for *HEIF*).
- **To get JPGs from the iPhone instead,** on the iPhone open **Settings > Photos** and, under *Transfer to Mac or PC*, choose **Automatic**. The iPhone then converts photos to JPG as they're copied. (**Keep Originals** copies the HEIC files.)
- **To find the HEIC files in the archive:** `da db query "SELECT path FROM files WHERE extension='.heic'"` .
- **Coming soon:** a DA command that makes JPG copies of HEIC photos, e.g. for sharing.

Edit an archive's settings (da-archive.yaml)

Each archive's settings are in `da-archive.yaml` in its folder.

1. **Make a copy first:** right-click `da-archive.yaml` > **Copy**, then **Paste** in the same folder. If anything goes wrong, rename the copy back.
2. **Open it with Notepad** (right-click > **Open with** > **Notepad**).
3. **Change what you need.** Keep the spaces at the start of each line exactly as they are; they matter. Lines starting with `#` are notes and are ignored. ```yaml archive: name: "jane-archive-199x" # the name shown at the top of the menu years: "1990-1999" # the years this archive takes; null for any year

   intake: # the folders in intake\ and what each one holds photos: photo docs: document scans: document receipts: document # an extra folder: anything in it is filed as a document ``` 4. **Save, then check:** open the

archive's menu. If DA can't read the file, it says which line is wrong; fix it or put the copy back.

A new intake folder you add here is created the next time you run `da setup` on the archive (`da setup D:\jane-archive-199x`), or you can create it yourself.

Share an album with Google Photos

DA doesn't upload to Google (Google no longer lets apps like DA create shared albums), but your filed photos are easy to share by hand:

1. Go to photos.google.com and sign in.
2. Click **Upload** (or drag photos onto the page) and pick them from the archive, e.g. `D:\jane-archive-199x\photos\organized\1998\1998-07\` .
3. Click **Create** (the **+**) > **Album**, give it a name, and add the uploaded photos.
4. In the album, click **Share** and choose people, or **Create link** to send the link to anyone.

The originals stay in your archive; Google gets copies.

Process a folder from somewhere else

For photos on a USB stick, an old backup drive, or anywhere outside the archive:

```
da run E:\old-backup\photos --archive D:\jane-archive-199x
```

DA offers to **copy** the folder into the archive's `intake\` and processes the copy, so it can check every photo against what's already filed. The originals aren't touched; delete them yourself once you're happy.

Start a new archive for another person or decade

```
da setup D:\sam-archive-201x --years 2010-2019
```

Then double-click its `Open DA.cmd` . To make an archive from a folder that already has photos in it, point `da setup` at that folder: DA moves its contents into `intake\` , and lists every move in `output\runs\<date>-setup\moved.csv` (where each item was, and where it went).

Use DA on another computer

An archive is just a folder, so it can live on a USB drive and move between computers.

1. Install DA on the other computer.

2. Plug in the drive and double-click the archive's `Open DA.cmd`. It works whatever drive letter Windows gives the drive.
3. Each computer keeps its own list of archives for **6 Switch archive**. Opening an archive once with its `Open DA.cmd` adds it to that computer's list.

Back up an archive

Everything about an archive is inside its folder: the photos, the documents, the settings, the sidecar notes and the records. To back it up, copy the **whole archive folder** to another drive (File Explorer: right-click the folder > **Copy**, open the backup drive, **Paste**).

- `_duplicates-holding\` only holds copies; you can leave it out.
- The `.da\` folder holds DA's index. It's fine to copy it, and DA can rebuild it if it's missing (**5 Refresh index**).

Check what's already filed

```
da validate photos\organized
da validate documents\organized\2010
```

This checks names, folders and sidecar notes, and finds duplicates inside that folder. It's a check only: nothing moves (it just adds any missing sidecar notes). The report is saved in `output\validateit\`.

To fix what it found in `organized` (wrong names, a photo in the wrong month), add `--mode repair`. On intake, or on a folder outside the archive, repair stops and tells you to use da run instead, which sets duplicates aside first.

Do the run's steps one at a time

The same steps as da run, one command each, for when you want to look closely between them:

```
da db build                                          # 1 index
da intake hash --folder phone-2024                   # 2 hash (folder inside intake)
da intake set-aside --folder phone-2024              # 3 plan only: what would be set aside
da check output\runs\<ts>-set-aside\plan.csv         #   re-check a sample of the plan
da intake set-aside --folder phone-2024 --apply      #   set them aside
da validate intake\phone-2024 --kind photo           # 4 preview
da check output\validateit\validateit-report-<ts>.json   # 5 check
da validate intake\phone-2024 --kind photo --mode repair --force   # 6 file (--force: you did 2
da verify <the repair report it printed>
da db build                                          # 7 refresh
```

Coming soon

- **Contact sheets:** a PDF of small pictures of every photo in a folder, made automatically when DA files photos, and kept up to date.
- **Live Photo pairing:** each iPhone `.MOV` filed next to its photo, and the ones left behind found.
- **HEIC to JPG:** JPG copies of iPhone photos for sharing.
- **A tag command:** `da tag` instead of editing sidecars by hand.

If something goes wrong

What you see	What to do
'da' is not recognized	Close the terminal and open a **new** one. If that doesn't help, install DA again.
Windows protected your PC	Click **More info**, then **Run anyway**. DA isn't signed by a company, so Windows asks once.
Not inside an archive	Open the archive with its `Open DA.cmd`, or type `da` and pick one, or add `--archive D:\jane-archive-199x` to the command.
DA.exe is in use while updating	Close every DA window, then install again.
A run seems stuck	It's probably working through a big folder. Big runs take hours; the step number shows where it is. To stop, press **Ctrl+C**. Photos already filed stay filed, and running it again carries on with the rest.
A photo was filed with the wrong date	The camera's clock was wrong. Rename the file in `organized` with the right date at the start, and move it to the right month folder.
Nothing to file	Everything in that folder is either filed or waiting for you. See **3 What needs review**.
Files are in `_duplicates-holding` that you want back	**4 Undo last set-aside**, or `da intake undo "<run folder>\set-aside-moved.csv"`.

Reference

How DA works out a date

In this order:

1. A date **and time** in the file name (`2024_07_04_10_15_00.jpg` , `20240704_101500.jpg`).
2. The date stored **inside** the file:
   - photos: the camera's "date taken", including iPhone HEIC photos. A camera whose clock was never set (1 January 1970) doesn't count.

    - PDFs: when the PDF was made (scanned, printed or saved).
    - Word, Excel, PowerPoint: when the document was created.
3. A date **without a time** in the name (`2010-04-27` , or US-style `04-27-2010`). If the date inside the file is from that same day, its time is used too.
4. A **month and year** in the name (`Resume April-2010`): the date inside the file, if it's from that month. Otherwise only the month is known, which isn't enough, and the file stays for review.
5. A **year only** (`Beach 1998.jpg`): not enough; it stays for review.

When the name and the date inside disagree, the name wins, and the file is listed for review.

Photos are named `photo_YYYYMMDD_HHMMSS_xxxx.jpg` and documents `doc_YYMMDD_HHMMSS_xxxx.pdf` ; the last four characters come from the file's fingerprint, so two photos taken in the same second still get different names.

Photo or document?

PDFs and Word, Excel and PowerPoint files are always documents. A picture (`.jpg .png .tif .bmp`) could be a photo or a scan, so the intake folder decides: in `docs\` or `scans\` it's a document; anywhere else it's a photo. To decide for a whole run: `da run intake\old-scans --kind document` .

Which archive a command works on

1. The one you name: `--archive D:\jane-archive-199x` (before or after the command).
2. Otherwise, the archive the folder you name is in.
3. Otherwise, the archive you're in: the terminal's current folder, or one above it.

Commands

Command	What it does
`da`	The menu
`da run [folder]`	File what's in intake (or one folder of it)
`da status`	This archive at a glance
`da setup <folder> [--years 1990-1999]`	Create an archive, or turn a folder into one
`da archives [--open N]`	The archives on this computer
`da guide [--pdf]`	Open this guide (or its PDF)
`da db build` / `da db stats` / `da db query "<SQL>"`	The index: refresh, counts by year, search
`da intake status` / `da intake undo <moved.csv>`	Intake progress; put duplicates back
`da validate <folder>`	Check a folder (nothing moves)
`da findit tags <folder> -r <tag>`	Find files by tag
`da version`	Which version of DA this is
`<DA.exe> install`	Install or update DA on this computer